

Technical Document #12: Natural Language Processing - Comprehensive Overview

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Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Question answering systems extract answers from a given context.
- Word embeddings represent words as dense vectors in continuous space.
- Tokenization is the process of breaking text into individual units called tokens.